Ex-01: Study of Various commands used in Linux and Windows

Networking commands:

* arp -a → Displays ARP table (IP ↔ MAC address)
* hostname → Shows the computer name
* ipconfig → displays complete ip configuration
* nbtstat -a → checks Net Bios name resolution
* nslookup → performs DNS lookup (finds IP address of a domain)
* pathping → combines ping + trace route with packet loss stastics
* ping → Test connectivity between two devices using icmp.
* route → displays or modify the routing table.

Linux Networking command:

1) ip: modern command for network config

common uses

* Show ip address
* Add / Delete ip address
* Bring interface up / Down
* Add / Delete routes
* View routing information.

2. ifconfig:

* Older command for viewing and configuring network interface.
* Mostly replaced by ip command.

3. mtr:

* Combines ping + Trace route
* Shows route, latency and packet loss
* Useful for network troubleshooting.

4. tcpdump:

* Captures and analyzes network packets
* Can filter traffic by:
  → interface → Host → Network

5. ping:

* Test connectivity between systems
* Measure response time
* Helps detect packet loss

nmcli (Network manager CLI): Configure ethernet connection

Troubleshooting Checklist:

* Check network enable
* Verify switch connections
* Ensure network interface is working.

Result:

Got output and Verified

Ex-02: Assign IP address manually to the Computer

Aim:

To assign ip address manually to the computer

Steps to assign:

1. View existing connections
   - nmcli connection show
2. Set a static ip v4 address
   - nmcli connection modify "wired conn-1"/
     ipv4.method manual/
     ipv4.address 192.0.2.1/24/
     ipv4.gateway 192.0.2.254\
     ipv4-dns 192.0,200/

   * ip address: 192.0.2.1/24
   * gateway: 192.0.2.254
   * dns server: 192.0.2.200
3. Activate the connection
   - nmcli connection up "wired conn-1"
4. Verify the ip address
   - ip address show
5. Verify the default Gateway
   - ip route show
6. Test the Network
   - ping <host_name-or-ip-address>

Result: 27/11/25

Hot output and Verified.

EX-03: Network cables and Ethernet Cables making

Types of Network Cables:

i) UTP - Unshielded Twisted Pair

* Cat 3 → 10 Mbps
* Cat 5 → upto 100 Mbps
* Cat 5e → 1 Gbps
* Advantages: Cheap, easy to install
* Disadvantages: more affected by EMI/noise
* Uses: Ethernet networks

ii) STP - Shielded Twisted Pairs

* Cat 6/6a → upto 10 Gbps
* Advantages: Shilded, faster that UTP
* Disadvantages: Expensive, more installation effort
* Uses: Gigabit, data centres.

iii) SSTP -

* Cat 7 → upto 10 Gbps
* Advantages: Better Shieding and reduced interface
* Disadvantages: Expensive & difficult to install

iv) Coaxial Cables:

* Types: RG-6, RG-59, RG-11
* Speed: 10-100 Mbps
* Uses: TV networks and high speed Internet connections.

v) Fibre Optic Cable:

* Type: Single-mode and multi mode
* Speed: upto 100 Gpbs
* Advantages: High Speed, High bandwidth, Secure
* Disadvantages: Expensive and requires skilled installation

Making ethernet Straight-Trough / Crossover Cable:

Tools and materials:

* Cat 5e ethernet Cable
* RJ45 connectors x2
* Crimping tool
* Cable Stripper
* Optional Cable Tester

Steps:

Step 01: Prepare the Cable
Step 02: Untwist the wires
Step 03: Arrange the wires
Step 04: insert wires into RJ45
Step 05: Crimp the connector
Step 06: Complete the Cable
Step 07: Test.

EX:04 SETUP and configure a LAN using a Switch and Ethernet cables in you Lab

Aim:

To setup and configure a Local Area Network using a switch and ethernet cables, assign IP addresses verify connectivity, and enable file sharing.

Procedure:

1. Planned the LAN topology with 4 PCs, 1 Switch, and ethernet cables
2. Connected all computers to the switch using ethernet cables
3. Assigned static IP addresses to each PC:

PC1 - 10.1.1.1

PC2 - 10.1.1.2

PC3 - 10.1.1.3

PC4 - 10.1.1.4

Subnet Mask - 255.0.0.0

4. Configured the switch with IP address 10.1.15 and Subnet mask 255.0.0.0
5. Verified network connectivity using the ping command
6. Shared a folder on one PC and accessed it from other computers on the LAN.

IP Configuration:

Device	IP Address	Subnet mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0
Switch	10.1.1.5	255.0.0.0

Result:

A LAN was successfully configured using a Switch and ethernet cables. All computers communicated successfully through the ping command, and shared folders were accessible from other systems on the network.

EX-NO: 05 Design a simple topology and configure with one router, two switches and four PC's in cisco packet tracer.

Aim:

To design and configure a simple network, topology using one router, two switches and 4 pc's in cisco packet tracer, configure static, configure static IPV4 addressing and router interfaces, and verify communication within and between two LAN using ping command.

Procedure:

1) Open Cisco packet Tracer & create a network topology by placing one router, two switches & four PC's in logical workplace.

2) Connect the devices using Copper Straight through

- PC0 → Switch0 (East ethernet 0/1)
- PC1 → Switch0 (Fast ethernet 0/2)
- PC2 → Switch 1 (Fast ethernet 0/1)
- PC3 → Switch 1 (Fast ethernet 0/2)
- Switch 0 (Gigabit ethernet 0/1) → Router 0
- Switch 1 (Gigabit ethernet 0/1) → Router 0

3) Configure the PC's with the following static IPV4 addresses:

- PC0: 192.168.10.10/24 Default Gateway: 192.168.10.1
- PC1: 192.168.10.11/24 Default Gateway: 192.188.10.1
- PC2: 192.168.20.10/24 Default Gateway: 192.168.20.1
- PC3: 192.168.20.11/24 Default Gateway: 192.168.20.1

4) Open the router CLI and configure the router interfaces :

* Assign 192.168.10.1/24 to Gigabitethernet 0/0 and enable it using no Shutdown.
* Assign 192.168.20.1/24 to Gigabitethernet 0/1 and enable it using no shutdown.

5) Save the router configuration using the write memory command.

6) Verify all physical links change from red to green, indicating that the interfaces are operational

7) Test Connectivity by :

* Pinging the default Gateway (192.168.10.1) from PC0.
* Pinging PC2 (192.168.20.10) from PC0 to verify inter-subnet communication.

8) Observe the ping results and Confirm successful communication between the two LAN's

Result:

The network topology was successfully configured in Cisco packet Tracer. All devices communicated successfully, and connectivity b/w both LAN's were verified using ping command.

Ex-NO-06 : Develop a customized ping command to test the Server connectivity.

Aim:

To develop a customized ping command in python to check the connectivity between a client & Server by sending ICMP echo request packets and displaying the response time and Status.

Requirements:

* Computer / laptop
* Internet connections
* Python 3.10 or above
* vs code / Pycharm
* Command prompt

Theory:

Ping is a network diagnostic tool used to determine whether a host is reachable over a network. It works using the ICMP protocol. The customized ping program accepts a hostname or IP address, sends multiple ping requests, displays response time and packet loss, and indicates the server status.

Procedure:

1. Install python and vs code
2. Create a python file named ping.py
3. Import the required models
4. Enter the hostname or IP address

5. Detect the operating system.
6. Execute the appropriate ping command
7. Display the ping results and packet stastics.
8. Check the return code and display the status

Program Code:

```
import subprocess
import platform
host = input("Enter website or IP address:")
System = platform.System()
if system == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(
        command,
        capture-output = True,
        text = True
    )
    print("\n ping results:\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server Status: Reachable")
    else:
        print("Server Status: Unreachable")
except Exception as e:
    print("Error:", e)
```

Output:

Enter website or IP address: google.com

Ping results:

pinging google.com [142.251.221.142] with 32 byte:

Reply from 142.251.221.142: bytes=32 time=3ms

Reply from 142.251.221.142: bytes=32 time=3ms

Server status: Reachable.

Result:

The customized ping command was successfully developed and executed using python.

EX-01: CISCO PACKET TRACER (SIMULATION TOOL)

AIM:

To Study the installation and user interface of cisco Packet Tracer and understand its environment for designing simple networks.

COMPONENTS:

1. Menu Bar
2. Main Toolbar
3. Logical / Physical Workspace
4. Workspace
5. Common Tools Bar
6. Realtime / Simulation Tabs
7. User - Created Packet Box

PROCEDURE:

1. Open CISCO packet Tracer
2. Place 4 PCs and 1 Hub in the Workspace
3. Place another PCs and 1 Switch
4. Assign IP addresses and Subnet masks to PCs
5. Use the PDU / messages tool to Send or find a packet from one PC to another
6. Repeat the process and compare how hub and Switch forwards Packets

RESULT;

CISCO packet Tracer in simulation mode is Studied successfully.

EX.NO - 08 : Customize Switch with Network module using Cisco packet Tracer

AIM:

To customize a modular switch in Cisco packet Tracer by installing Fast ethernet and Gigabit ethernet fiber network modules and to establish and verify connectivity with PCs and a CISCO 2911 router

PROCEDURE:

1. Cisco packet Tracer was opened.
2. Switch PT - empty was selected from Network Device
3. two PCs and one CISCO 2911 router were added
4. The Physical tab of the switch was opened
5. Two PT - switch - NM - IFEE module → Slot 0 & 1
6. Switch powered ON
7. The CLI was opend and the Switch boot process
8. PC-0 was connected to Fast ethernet 0/1
9. PC-1 was connected to Fast ethernet 1/1
10. Router is powered ON
11. The Router interface Gigabit Ethernet 0/0/0
12. IP address set to 192.168.10.1 and subnet mask 255.255.255.0
13. The no shutdown command was used to activate the router interface.
14. Connectivity was tested using the ping command

COMMAND USED:

```
Switch > enable
Switch Show up interface brief
Router > enable
Router # configure terminal
Router (Config ) # Interface Gigabit Ethernet 0/0/0
Router ( config )# ip address 192.168.10.1 255.255.255.0
Router (config -if )# exit
Router (Config ) # exit
Router # write memory
Router # Show ip interface brief
Router # Show interfaces Gigabit ethernet 0/0/0
Router # ping 192.168.10.2
```

RESULT:

Thus, the modular switch was successfully customized using appropriate network modules in cisco packet Tracer.

EX.NO-09: Building an anonymous FTP Scanner using python

AIM:

To develop a python-based anonymous FTP Scanner using the build-in ftplib module and determine whether an FTP server permits anonymous

REQUIREMENTS:

- Python 3.x
- Python build-in ftplib module
- Authorized FTP server for testing
- Internet / Network Connection

PROCEDURE:

1. Install Python 3.x on the computer
2. Open command prompt or terminal
3. Create a python file named anonymous-ftp-scanner.py
4. Import the FTP all-cross functions
5. Define the anonymous FTP_scan() function
6. Enter the authorized FTP server add
7. Connect to the FTP server using port 21
8. Test anonymous login using username anonymous
9. Display the accessible directory display contents using nlst(). Close the FTP connection and display the result.

RESULT:

It successfully determined whether FTP server is displayed.

EX.NO - 10 : Implement packet sniffing using RAW Sockets in python

AIM:

To implement a packet sniffer RAW sockets in python and capture network packets by displaying important if header information such as soure IP, distination IP, protocol, TTL and packet length.

PROCEDURE :

1. Install python and add it to the system PATH.
2. Create a project folder named packet sniffer
3. Creat a python file named sniffer.py
4. Creat a RAW sockets using socket.AF-INET
5. Enable IP header inclusion and promissoous mode.
6. Extract and decode the IP header using Struct unpack()
7. Displaces the IP version, header length, TTL, protocol, Source IP, distination IP and packet length.
8. Stop the packet Snifer using Ctrl+C

COMMANDS USED:

Python --version
Python
cd desktop

cd packet sniffer
python sniffer.py
Sudo python sniffer.py

- These are the commands that used.

RESULT:

The packet sniffing program was succesfully implemented using RAW sockets in python.

EX NO: 11 Nmap Live Host Discovery

AIM: To discover live hosts on network using Nmap host-discovery scans such as ARP, ICMP, TCP and UDP Ping scans, and to understand the protocol used to each layer of the TCP/IP model.

PROCEDURE:

1. log in to TryHackMe and open the Nmap Live Host Discovery
2. Start the target network and launch the AttackBox.
3. Open the terminal in the AttackBox.
4. Note the target / connection - IP
5. Perform ARP -PR
6. Perform ICMP Echo using -PE, -PP and -PM
7. Perform TCP SYN & TCP ACK -PS and -PA
8. Perform UDP Ping discovery using -PU

COMMANDS USED:

```
Sudo nmap -PR -sn 10.200.6.0/24
Sudo nmap- PE -sn 10.200.6.0/24
nmap -PP -sn 10.200 6.0/24
nmap -PM -sn 10.200.6.0/24
nmap- PS -sn 10.200.6.0/24
Sudo nmap -PA -sn 10.200 6.0/24
sudo nmap -PU -sn 10.200.6.0/24
```

RESULTS: The Nmap live host discovery experiment was successfully performed.

EX NO: 12 Develop a simple calculator using XMLRPC

AIM: To design, implement, and verify a remote calculator application using the XML-RPC protocol in python.

PROCEDURE:

Server side:

1. import SimpleXMLRPCServer from the xmlrpc.server
2. Define functions for addition, subtraction, division.
3. Add a check to prevent division by zero
4. Create the XML-RPC server on localhost at port 8000.
5. Start the server using serve_forever()

Client side

1. Import the xmlrpc.client module
2. Create a connection to http://localhost:8000/ using
3. get two floating-point numbers from the user

COMMANDS USED:

Python Calciserver.py
Python Calciclient.py
Calciserver.py
Calciclient.py

OBSERVATION:

Addition = 30.0
Subtraction = 10.0
Multiplication = 200.0
Division = 20

RESULT: The XML-RPC based simple calculator application was successfully developed and executed.